Anonymous Submission (Double-Blind Review)

Geometric Computability on Grothendieck Topoi: The Abstract Tensor Machine (POLYDIM V2)

Abstract

This paper introduces the mathematical formalism of POLYDIM V2, establishing the paradigm of **Geometric Computability on Grothendieck Topoi (GCGT)**. We address the current neuro-symbolic crisis in artificial intelligence, where discrete logical reasoning and continuous tensor-based optimization coexist fragilely. By formalizing the hyperdimensional latent space as a Grothendieck Topos, we demonstrate that logical and safety invariants do not require external algorithmic validation, but instead act as purely spatial constraints (coverage sieves). We synthesize 15 mathematical pillars, including Lurie's ∞-categories, Clifford Algebra (Multivectors), Directed Topology (Ditopology), and Combinatorial Hodge Theory, to define an AI architecture that is resilient to floating-point degradation, immune to causal paradoxes, and natively parallel.

1. Introduction: The Failure of the Euclidean-Symbolic Paradigm

The current deep learning architecture (Transformers, LLMs) models representation spaces as Riemannian manifolds, evaluating semantic similarities via the dot product in $\mathbb{R}^N$. While this Euclidean model captures complex statistical distributions, it suffers from a fatal weakness: it lacks strict formal semantics. The system does not reason logically; it approximates geometrically. Any attempt to force standardized logical rules (imperative code, schema validation) onto these continuous vectors requires an external "bridge", resulting in hallucinations when geometry diverges from logic, or critical failures in autonomous agent architectures. POLYDIM V2 solves this neuro-symbolic dichotomy by redefining computation itself.

2. The Latent Space as a Grothendieck Topos

Instead of keeping logic and geometry separate, POLYDIM V2 operates on a

Grothendieck Topos.

2.1 The Site and Semantic Coverage

Let $\mathcal{C}$ be a base category where objects are semantic subspaces. We endow $\mathcal{C}$ with a Grothendieck Topology J. A logical proposition (e.g., a database validation) is formalized as a sheaf over this site. The Subobject Classifier Ω determines truth not as a boolean ${0,1}$, but as a topological patch. An invalid state results in a pullback to the empty set ($\emptyset$). Logical constraints are computed at the speed of topology.

2.2 Internal Intuitionistic Logic

The system operates under Heyting Logic. The absence of the law of excluded middle allows latent states to flow smoothly across semantic boundaries (superposition) without violating macroscopic logical consistency.

3. Geometric Dynamics: From Matrices to Rotors

3.1 Clifford Algebra (Multivectors)

Using flat weight matrices $U \cdot V^T$ causes distortions. POLYDIM V2 expands to the algebra $C\ell_n(\mathbb{R})$. States are multivectors (combinations of points, lines, areas, volumes). The superposition operator `MIX` is formalized as a Spin **Rotor**, performing pure continuous rotations ($R = e^{\theta B/2}$) that preserve latent semantic volume and avoid the gimbal lock of high dimensionality.

3.2 The PROJECT Operator and Hodge Duality

A subspace is a "Blade" I_k. Projection is $\text{Project}_{I_k}(M) = (M \cdot I_k) I_k^{-1}$. Navigating to orthogonal dimensions to avoid cognitive collapse is done in $O(1)$ using the Hodge Dual $M^* = M I^{-1}$.

4. Homotopic Stability and Resilience

4.1 ∞-Categories and Kan Complexes

Strict equality is impossible on SIMD hardware. We evaluate the `COMPOSE` composition in a Lurie quasi-category. If two computational paths differ due to precision errors, the identity (`GEO_ID`) is not destroyed. The system applies **Kan Horn Filling** to infer the 2-simplex (homotopy) that mathematically stabilizes both paths into the same equivalence volume.

4.2 Ditopology and Irreversible Causality

The infusion of **d-paths** restricts inference trajectories to pre-ordered directions. It is geometrically impossible for the network to experience "logical time loops" or backtrack into causally destroyed states.

5. Transport Flow and Harmonic Convergence

5.1 Information Geometry

The `MIX` operator does not sum vectors; it interpolates across Amari's α-connections on the Fisher-Rao metric. Convergence is calculated by optimizing the inverse of the Fisher Tensor (F^{-1}) via the Woodbury identity and KFAC in $ \mathcal{O}(d \cdot r)$, solving the computational bottleneck of the Natural Gradient.

5.2 Combinatorial Hodge Decomposition

To filter noise during the global state operator `FIXPOINT`, the system computes the Laplacian on cell complexes $\Delta_1 = B_2 B_2^T + B_1^T B_1$. Using the Hodge decomposition theorem, we isolate: - **Gradient:** Pure transitive flow. - **Curl:** Circular noise or attention stagnation. Monitored via *Rotational Energy* ($E_R(\omega)$). - **Harmonic:** The stable and safe convergence, representing the true response to computation.

6. Conclusion

The Abstract Tensor Machine V2 shifts artificial intelligence from statistical empiricism to universal mathematical rigor, enabling truly autonomous systems capable of unbreakable reasoning at the scale of massive parallel hardware.